REQUIREMENTS GATHERING & SPECIFICATION

Federal Workforce Intelligence & Retention Analytics

Comprehensive Business & Technical Requirements Framework

1. IDENTIFY STAKEHOLDERS

CORE **Project Team**

Responsible for overall data collection, processing, modeling, analysis, and building interactive analytics visuals.

REVIEW **Academic Supervisor**

Reviews project requirements, methodology rigorousness, technical architecture, and final analytical deliverables.

USER **Workforce Analysts**

Primary end-users requiring granularity in tracking overall workforce size, monthly hiring trends, and employee separations.

USER **Workforce Decision-Makers**

Strategic users utilizing intelligence and retention reports to support organizational staffing, policy, and budget planning.

2. PLAN THE APPROACH

> **Primary Objective:** Analyze large-scale federal workforce datasets sourced from the U.S. Office of Personnel Management (OPM), focusing on three major historical datasets: *Federal Employment*, *Federal Accessions*, and *Federal Separations*.

The team will inspect raw source files, review the OPM Data Dictionary, align fields across datasets, define strategic business questions, and formulate Key Performance Indicators (KPIs) through a structured 3-tier data architecture:

Bronze Layer

Raw ingestion of OPM TXT files into SQL Server without structural alterations.

Silver Layer

Cleaned, standardized, deduplicated, and formatted relational schemas.

Gold Layer

Aggregated business models, dimensional data marts, and KPI views.

3. CONDUCT ELICITATION

The project requirements are centered on delivering clear answers to five fundamental business questions:

- How does the total size of the federal workforce evolve and change over time?
- How many workforce accessions (hires) and separations (attrition) occur each month?
- Which specific federal agencies and occupational groups exhibit the most significant workforce shifts?
- Are monthly workforce accessions sufficient to offset ongoing separations?
- What key macro trends indicate potential long-term workforce planning and retention challenges?

4. DOCUMENT OBSERVATIONS & CONSTRAINTS

Category	Key Specifications & Observations
Functional Requirements	• Ingest & process large OPM TXT files using Python. • Profile data quality: flag missing values, duplicates, and formats. • Store structured data in SQL Server across Bronze, Silver, and Gold layers. • Compute core metrics: Net Change, Accession Rate, and Separation Rate. • Build interactive dashboards equipped with dynamic filtering controls.
Non-Functional Requirements	• Process multi-million record datasets efficiently. • Ensure end-to-end data accuracy, lineage, and audit consistency. • Maintain fully reproducible ETL code and documented execution steps. • Deliver user-friendly, high-performance executive dashboard views.
Project Constraints	• Strict **4-week project timeline**. • Memory-efficient processing required due to high file size. • Cross-dataset joining constrained by available OPM data dictionary fields.

5. ANALYZE & PRIORITIZE (MOSCOW FRAMEWORK)

MUST HAVE — P1

Python TXT ingestion pipelines, data cleaning routines, SQL Server DW architecture, core KPIs (Accessions, Separations, Net Change), and basic functional dashboard.

SHOULD HAVE — P2

Agency and occupational group comparisons, detailed multi-year trend analytics, and automated data-quality validation rules.

COULD HAVE — P3

Workforce forecasting algorithms and advanced statistical time-series analysis (subject to data cleanliness and schedule bandwidth).

WON'T HAVE — Out of Scope

Individual employee attrition prediction models in initial release, due to lack of reliable longitudinal employee IDs in public files.

6. VERIFICATION, VALIDATION & SCOPE MANAGEMENT

Verify & Validate

Verify data definitions against OPM dictionary standards. Check record counts, date ranges, and KPI logic. Conduct interim reviews with the Academic Supervisor before final submission.

Manage Scope Changes

Document all modifications to KPIs, datasets, or visual components. Strictly evaluate optional features against the rigid 4-week project deadline to eliminate schedule slippage.

7. EXPECTED FINAL DELIVERABLES

01. PYTHON	02. SQL DW	03. QUERIES	04. BI VIZ	05. REPORT
Ingestion & Profiling Scripts	**Bronze/Silver/Gold Warehouse**	**Validated KPI & Analytics Views**	**Power BI / Tableau Dashboards**	**Final Analytical Documentation**